

Multiple Choice (10 marks)

1. What is the significance of the 1:2:4 resonance in the Jupiter's moons system?
 - (a) The resonance pulls Io in different directions and generates heat.
 - (b) It makes the orbit of Io slightly elliptical.
 - (c) It creates a gap with no asteroids between the orbits.
 - (d) It prevents formation of the ring material into other moons.
2. Why do we see essentially the same face of the Moon at all times?
 - (a) because the Moon has a nearly circular orbit around the Earth
 - (b) because the Moon does not rotate
 - (c) because the other face points toward us only at new moon when we can't see the Moon
 - (d) because the Moon's rotational and orbital periods are equal
3. What drives differentiation?
 - (a) Spontaneous emission from radioactive atoms.
 - (b) The minimization of gravitational potential energy.
 - (c) Thermally induced collisions.
 - (d) Plate tectonics.
4. What are the conditions necessary for a terrestrial planet to have a strong magnetic field?
 - (a) fast rotation only
 - (b) a rocky mantle only
 - (c) a molten metallic core only
 - (d) both a molten metallic core and reasonably fast rotation
5. Which of the following lists the ingredients of the solar nebula from highest to lowest percentage of mass in the nebula?
 - (a) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) light gases (H He) metals rocks
 - (b) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) light gases (H He) rocks metals
 - (c) light gases (H He) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) metals rocks
 - (d) light gases (H He) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) rocks metals

True / False (10 marks)

Answer True or False

6. True or False: The black hole at the center of the Milky Way galaxy is known as Sagittarius A*.
7. True or False: A larger object, like a baked potato, retains heat longer than a smaller object due to its greater volume-to-surface area ratio.
8. True or False: Radioactive dating of lunar samples indicates that most lunar rocks are significantly older than their terrestrial counterparts.
9. True or False: When the Moon is setting at noon, it is in the third quarter phase, positioned 90 degrees behind the Sun in the sky.
10. True or False: Jupiter and other jovian planets contain a significant fraction of non-gaseous matter, despite being called 'gas giants.'

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the characteristics and significance of the Great Red Spot on Jupiter, including its formation, longevity, and impact on our understanding of atmospheric phenomena in planetary science.
12. Discuss the implications of the distance between Earth and Altair being one million times that of the Earth to the Sun, and calculate the exact distance from Earth to Altair.

End of Paper